

Experiment 2

AIM: Design Employee Payroll System using Inheritance, Interface & Polymorphism

Source: Practical 2.cs

Code:

```
using System;
namespace Practical2 {
    interface EmpSalary
    {
        double CalculateSalary();
    }
    class FtEmp : EmpSalary
    {
        public int empId;
        public double basicSalary;
        public double hra;
        public double da;

        public FtEmp()
        {
            Console.WriteLine("Full Time Employee object created");
        }
        public double CalculateSalary()
        {
            return basicSalary + hra + da;
        }
    }
    class PtEmp : EmpSalary
    {
        public int empId;
        public double basicSalary;
        public double hra;
        public double da;

        public PtEmp()
        {
            Console.WriteLine("Part Time Employee object created");
        }
        public double CalculateSalary()
        {
            return basicSalary + hra + da;
        }
    }
}
```

```
}  
class Program  
{  
    static void Main(string[] args)  
    {  
        Console.WriteLine("Enter Employee Type (F for Full Time, P for Part Time):");  
        string empType = Console.ReadLine().ToUpper();  
        if (empType == "F")  
        {  
            FtEmp ftEmp = new FtEmp();  
            Console.WriteLine("Enter Full Time Employee Details:");  
            Console.Write("Enter Employee ID: ");  
            ftEmp.empId = int.Parse(Console.ReadLine());  
            Console.Write("Basic Salary: ");  
            ftEmp.basicSalary = double.Parse(Console.ReadLine());  
            ftEmp.hra = ftEmp.basicSalary * 0.50;  
            ftEmp.da = ftEmp.basicSalary * 0.20;  
  
            Console.WriteLine($"Full Time Employee Salary: {ftEmp.CalculateSalary()}");  
        }  
        else if (empType == "P")  
        {  
            PtEmp ptEmp = new PtEmp();  
            Console.WriteLine("Enter Part Time Employee Details:");  
            Console.Write("Enter Employee ID:");  
            ptEmp.empId = int.Parse(Console.ReadLine());  
            Console.Write("Basic Salary: ");  
            ptEmp.basicSalary = double.Parse(Console.ReadLine());  
            ptEmp.hra = ptEmp.basicSalary * 0.30;  
            ptEmp.da = ptEmp.basicSalary * 0.10+;  
  
            Console.WriteLine($"Part Time Employee Salary: {ptEmp.CalculateSalary()}");  
        }  
        else  
        {  
            Console.WriteLine("Invalid Employee Type");  
        }  
        Console.ReadKey();  
    }  
}
```

Output:

```
C:\Users\student\Desktop\Net >
Enter Employee Type (F for Full Time, P for Part Time):
P
Part Time Employee object created
Enter Part Time Employee Details:
Enter Employee ID:14
Basic Salary: 58000
Part Time Employee Salary: 81200
|

C:\Users\student\Desktop\Net >
Enter Employee Type (F for Full Time, P for Part Time):
F
Full Time Employee object created
Enter Full Time Employee Details:
Enter Employee ID: 14
Basic Salary: 58000
Full Time Employee Salary: 98600
|
```

GitHub URL: <https://github.com/jainishkakkad133582/.Net-Practicals/tree/main>